

Automated Charles Schwab Trading Guide: Market Open Execution & Exit Lifecycle

System Architecture: Multi-Factor ML Momentum Swing Strategy

Account Profile: \$10,071 Live Capital | 1.5x Margin Leverage (\$15,106 Buying Power)

Risk Configuration: Ralph Vince Leverage Space Sizing | **8.0% Max Portfolio Heat Budget**

Position Constraints: Max 3 Concurrent Positions | Max 3 per Sector

1. What You Need to Run Right Now for Market Open Execution

To stage your trades so Charles Schwab executes them at the 9:30 AM ET market open:

Step A: Preview the Order Tickets (Dry Run)

Verify the ticket generation and allocation with your active 8.0% heat budget:

```
./auto_trade_schwab.sh --dry_run=true
```

Inspect the output file:

```
cat LATEST_TICKET.txt
```

You will see: - **Order Ticket #2: AMUU** (~10.8 shares, ~\$3,362 allocation) with GTC Stop-Loss at \$266.13, Tier 1 at \$378.77, Tier 2 at \$446.36. - **Order Ticket #3: PENG** (~46.3 shares, ~\$3,362 allocation) with GTC Stop-Loss at \$65.85, Tier 1 at \$82.75, Tier 2 at \$92.89. - **Cash Park Ticket: QQQ** (~4.3 shares, ~\$3,274 allocation) to earn index market yield on unallocated cash.

Step B: Submit Live Orders to Charles Schwab Desk

Run the live execution pipeline:

```
./auto_trade_schwab.sh --dry_run=false
```

What Happens Behind the Scenes at Schwab Overnight & Tomorrow Morning:

1. `execute_broker.py` connects to the Schwab API using your authenticated OAuth tokens.
 2. It transmits **parent market buy orders** with session settings `Duration: DAY` and `Session: NORMAL`.
 3. Because market hours are closed overnight, Schwab's order desk **queues the orders in pre-market staging**.
 4. At **9:30:00 AM ET tomorrow**, the opening bell rings and Schwab automatically fills the buy orders at the market opening print.
 5. Upon fill, Schwab's trading servers arm the **OCO (One-Cancels-Other) child bracket orders**:
 - The **GTC Hard Stop-Loss** is activated immediately on Schwab's servers (protecting downside 24/7).
 - The **Tier 1 Limit Sell (+1.5R)** is queued to take 50% profit automatically.
 - The **Tier 2 Limit Sell (+3.0R)** is queued to take 25% profit automatically.
 6. The fills are recorded in `portfolio.json`, locking in your purchase prices, stop levels, and entry timestamps.
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2. If Bought on Thursday, Does the Model Sell Next Thursday? (The 7-Day Rule)

Yes, strictly for non-runner lots that have not hit profit targets or stops.

The strategy uses **trading days (business days)**, not calendar days: - **Day 0 (Thursday)**: Entry fill at market open. - **Day 1 (Friday)**: 1st trading day completed. - **Day 2 (Monday)**: 2nd trading day completed. - **Day 3 (Tuesday)**: 3rd trading day completed. - **Day 4 (Wednesday)**: 4th trading day completed. - **Day 5 (Thursday)**: 5th trading day completed. - **Day 6 (Friday)**: 6th trading day completed. - **Day 7 (Monday of Week 3)**: 7th Trading Day (Time Expiration).

How Day 7 Expiration Works:

- If a trade has neither stopped out nor reached its Tier 2 target after 7 full trading sessions, its momentum edge has decayed.
 - At **3:30 PM ET on Day 7**, `daily_signal.R` flags the position:
Action: SELL (MAX 7-DAY TIME HORIZON)
 - The position is liquidated at market close, freeing up cash for fresh leaderboard candidates.
 - **Crucial Exception (The 25% Runner Lot)**: If the trade hit Tier 1 and Tier 2 targets earlier, the remaining **25% runner portion is 100% exempt from the 7-day expiration**. It is held indefinitely as long as it stays above its dynamic 2.5x ATR Chandelier trailing stop.
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3. What Happens Every Day When the Cron Job Runs the Scanner?

The daily cron schedule is set to run Monday through Friday at **3:30 PM ET (15:30)** via:

```
30 15 * * 1-5 /home/cmejo/Development/R_SwingTrading/auto_trade_schwab.sh --dry_run=false >> /home/cmej
```

When the cron fires at 3:30 PM each day, it performs **Portfolio Synchronization & Capital Deployment**:

[Cron 15:30 ET]

1. Sync Active Portfolio (``portfolio.json``)
 - Checks Schwab fills and current market prices.
 - Calculates elapsed trading days for each open position.
 - Updates trailing stops: ratchets stop to Breakeven (+1R) or Chandelier trailing stop.
 2. Check Exit Conditions
 - Did any position hit its Stop-Loss, Tier 1 (+1.5R), Tier 2 (+3.0R), or Day 7 Expiry?
 - If Friday: runs Friday Conditional Weekend Defense.
 - Generates SELL orders for any triggered exits before the 4:00 PM close.
 3. Check Available Cash Slots
 - Are any of the 3 slots empty?
 - IF ALL 3 SLOTS ARE OCCUPIED: The scanner outputs "0 open slots", leaves positions alone, and makes NO new purchases (no churning or overtrading).
 - IF 1 OR MORE SLOTS ARE EMPTY: Scans the 170-stock universe, selects the top candidates, calculates Ralph Vince sizing (with 8.0% heat cap), and stages BUY tickets for tomorrow.
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4. What Causes Selling on Friday? (The Weekend Risk Engine)

Every Friday at 3:30 PM ET (`is_friday = TRUE`), the model evaluates open positions against the **Conditional Weekend Holding Engine** in `R/portfolio_manager.R`.

Over a 48-hour weekend, the market is closed, leaving positions vulnerable to geopolitical news, macro shocks, or earnings announcements that could cause severe gap-downs on Monday morning.

The 5 Friday Weekend Tests:

A position is **APPROVED TO HOLD OVER THE WEEKEND** if and only if it passes all 5 criteria: 1. **Macro Gate**: QQQ 50-day moving average trend is BULLISH ($C1 > 1mMA50$). 2. **Model Probability**: ML Probability of upward continuation remains strong: $P(\text{Up}) \geq 55\%$. 3. **Weekly Synergy**: Weekly trend slope remains positive ($\text{Weekly_Slope} > 0$). 4. **Earnings Blackout**: No corporate earnings announcement scheduled within the next 7 calendar days. 5. **Stop Buffer**: Price is currently trading comfortably above the stop-loss level.

What Triggers an Immediate Friday Sell?

If an open position fails **even one** of these tests:

Action: SELL BEFORE 16:00 (Weekend Risk: [Failure Reasons])

The system triggers a market sell order between 3:30 PM and 4:00 PM on Friday to liquidate the position to cash, completely eliminating weekend gap risk.

5. If Not Sold Friday, Does the Model Always Hold for 7 Days?

No. Positions can sell earlier at any time if target or stop triggers hit.

Here is the exact hierarchy of how and when momentum trades exit:

Exit Trigger	When It Happens	Portion Sold	Action Taken
1. Hard Stop-Loss	Any time intraday (armed on Schwab servers)	100%	Full exit immediately to protect capital.
2. Tier 1 Target (+1.5R)	Day 1 to Day 6 intraday	50%	Sells 50% lot; moves stop on remainder to Breakeven.
3. Tier 2 Target (+3.0R)	Day 1 to Day 6 intraday	25%	Sells 25% lot; converts remaining 25% to Runner Lot.
4. Friday Defensive Exit	Friday 3:30 PM ET	100%	Liquidates positions that lost momentum or face earnings.
5. 7-Day Time Expiration	Day 7 at 3:30 PM ET	Remainder	Closes non-runner shares whose momentum expired.
6. Chandelier Runner Stop	Days 7 to 60+ (multi-week trends)	25%	Trails highest price by $2.5 \times \text{ATR}$; exits only on trend break.

6. Will the Model Run Completely on Its Own Without You?

Yes, the pipeline is fully autonomous end-to-end once your cron and Schwab tokens are active.

The Autonomous Daily Cycle:

1. **Automated Account Sync:** Charles Schwab cash and portfolio equity are synced automatically at the start of each run.
2. **Automated Order Staging:** Buy tickets are calculated and sent directly to Schwab's API.
3. **Automated Bracket Placement:** Schwab's server hosts the bracket orders, so stops and profit targets fill automatically even if your computer is asleep.
4. **Automated Position Tracking:** `portfolio.json` records every fill, counts holding days, and ratchets trailing stops without manual input.
5. **Automated Alerts:** Mobile webhook notifications (Email, Discord, or Telegram) notify you whenever orders execute or stops adjust.

Your Only Maintenance Tasks:

- **Weekly Token Refresh:** Charles Schwab OAuth refresh tokens expire every 7 days. Ensure your token refresh script runs weekly or re-authenticate via `./execute_broker.py --broker=schwab` if credentials expire.
- **System Monitoring:** Periodically check `cat LATEST_TICKET.txt` or review logs in `logs/` to confirm your cron job executed on schedule.